The Text Beside the Image: Detection, Utility and Leakage for Trustworthy Multimodal Medical Data and Beyond

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Abstract. Medical images are released with the reports that describe them, and protecting the image does not protect the report. This paper measures the text component of such releases. We measure identifier detection, downstream utility and residual identity leakage on the same documents, with the pseudonymisation policy as the variable under test: 15 detectors, three release conditions and four corpora of medical reports, legal judgments, news and other genres, and e-mail, in German, English, Chinese and Arabic. Across CARDIO:DE, TAB, OntoNotes and Enron, the same fixed 13-detector ensemble yields person sensitivity ≥ 0.9352 at specificity 0.6235–0.9318. With this ensemble, frequency matching with a public name list recovers zero identities by alignment across the four corpora; the names it got right were ones the detector missed, left in clear text. Cross-document linkage ranks the correct person first for 0.93 % of e-mail queries without training and 3.94 % with it, against 1/3697 chance and 71.98 % on unmodified text. On CARDIO:DE, linkage recovers zero without training and 0.71 % with training. These rates are in line with corpus properties.

Keywords: pseudonymisation · clinical text · text in multimodal releases · privacy evaluation

1 Introduction

A de-identified medical image does not protect the patient when its accompanying report retains identifiers. Artificial intelligence scales both tasks, and its errors vary by distribution, language, script and identifier class. Packhäuser et al. [28] re-identified patients from de-identified chest radiographs and priced a defence against diagnostic utility [29]; pathological speech has been evaluated against privacy, utility and fairness [34]; clinical text is instead kept with the model inside the hospital [13]. We ask what survives transformation of the text, and what protecting it costs.

De-identification reduces identifying information. Reversible *pseudonymisation* replaces an identifier with a realistic *surrogate*; *anonymisation* protects

Table 1: The corpora separate domain, language, and where the reference annotation came from.

Corpus	Docs	Tokens	Ment./doc	Lang.	Reference annotation
CARDIO:DE	400	885 059	78.3	de	constructed
TAB	1268	1 844 804	81.2	en	manual
OntoNotes	5994	4 726 594	45.5	en/zh/ar	linguistic
Enron	58 636	15 507 925	12.2	en	header-derived

individuals. Stable pseudonyms preserve longitudinal utility but hand an adversary a persistent handle. European guidance regards keyed message authentication codes as robust [11], and the health-informatics standard is revised accordingly [9].

Detection benchmarks ask whether an identifying span was found [15, 36, 38]; utility studies measure downstream change [2, 37] or information loss [35]; re-identification studies measure attack success. Frameworks combine technical performance, information loss and intrusion tests [21], while surrogate systems and masking benchmarks cover single pieces [10, 27, 31]. No study states together what one policy buys and what it costs. Data-protection officers release documents and cases, not average tokens, as one unchanged mention can identify someone after thousands were removed, so we measure surviving findings, exposed cases and linkable entities rather than token rates alone.

Our design covers four corpora in German, English, Chinese and Arabic: clinical letters, legal judgments, news and other genres, and e-mail. Every document receives the same detector comparison, release conditions and measurements. Two predictions are under investigation: that detector precision governs utility more strongly than the choice between surrogates and typed placeholders, and that detection sensitivity moves the two attacks in opposite directions, because removing a name obstructs context linkage while leaving a larger, more stable population for frequency matching.

2 Material and Methods

2.1 Corpora

Reference provenance differs across the corpora, and with it the detection task (Table 1). CARDIO:DE is the clinical anchor. To use it here, we inserted gold identifiers into an already de-identified release, which makes its name distribution synthetic. TAB manually annotates identifiers and document-scoped co-reference in court judgments [31]; a *named entity* is a predefined person, organisation, location or other type, its occurrences form a co-reference chain. OntoNotes covers three languages but was annotated for linguistics rather than privacy. Enron derives genuine cross-document identity from message headers but contains exposed personal information [25].

Therefore CARDIO:DE ground truth covers only inserted identifiers, and 84.7 % of its gold tokens are dates, which the detectors do find but the conditions

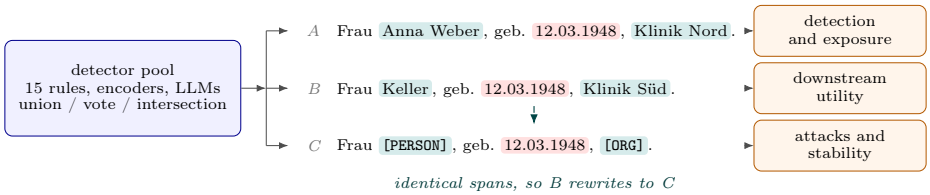


Fig. 1: One set of detector spans generates three release conditions, on an invented header line in the style of a discharge letter. Detected identifiers (teal) become surrogates in B and typed placeholders in C; dates (red) pass through by design. A surrogate replaces a whole detected span with a single name, so the two-token name becomes one. B and C replace the same spans, so C is a rewrite of B. Every condition meets every measurement family on the same documents.

pass through by design. Sensitivity, replacement, exposure and the attacks are therefore all scored over the types actually replaced, and the retained types are not part of our evaluation. Retained dates are themselves a re-identification channel, easy to spot, and a release keeping them needs date shifting. A TAB applicant is usually named once but has a median of 17 later role mentions, and only Enron has natural cross-document identity.

2.2 Conditions

Condition A is unmodified text. In condition B, each detected identifier becomes a realistic *surrogate*, such as another name, stable across the corpus, indexed by HMAC-SHA256, a keyed one-way digest irreproducible without the secret. Condition C uses an unindexed placeholder such as `[PERSON]`. B and C differ only in replacement form, following European guidance and prior e-mail pseudonymisation [10, 11], and replace identical spans.

Before hashing, the key normaliser casefolds mentions, collapses whitespace and removes titles and punctuation, so *Dr. Weber* matches *weber* but *Weber* does not match *Kay Weber*. Unrelated people may therefore share a key while one person's initials and full name may not; Sect. 3.3 reports both failures. Figure 1 evaluates every condition on identical documents and measurements.

2.3 Detectors, Ensembles and Operating Points

The 15 frozen detectors comprise a rule engine [32], a German domain tagger [1,10], two zero-shot models [42], a multilingual NER encoder [7], two clinical de-identifiers [6, 16] and eight large language models [12, 14, 18–20, 26, 41]; the repository, we release with acceptance of the paper, lists every model, version and prompt. Cached spans are combined by union, intersection or at-least-k vote. Two rules recur below: the *permissive* rule is the union of all 15, the *precise* rule a majority, 8 of the 15. Leakage uses the configuration we recommend, a fixed 13-detector union; two large language models were excluded from it, because

preliminary runs identified them as one to two orders of magnitude slower per document without improving detection.

To explore ensembling, each corpus reports fast and maximum detector triplets for sensitivity, the share of replaced-type identifier tokens detected, and specificity, the share of all other tokens left untouched; for each parallel run we report slowest member of the ensemble. Our search covers singles, pairs and triples. Still for maximum performance, we recommended the 13-detector ensemble. Note that it does exceed the maximum of Table 2 most of the time.

2.4 Measurements

Detection is scored by token and by entity, an entity counting as protected only when every one of its mentions was replaced. Exposure uses document, case and entity denominators, because few leaked tokens can expose many people. Frozen models analyse all three conditions for named-entity agreement, medication extraction and section classification; agreement is span F1 between the outputs on original and transformed text, and the original-text score accompanies every comparison. Continuous pairs use the Wilcoxon signed-rank test, binary correctness McNemar's test, and Benjamini–Hochberg correction applies within task families. Utility and attack figures carry one standard deviation over documents, detector combinations or folds; detection and exposure are exact population proportions.

A *collision* gives distinct people one surrogate, *fragmentation* gives one person several within a document, and *drift* changes it between documents. These apply to B, which has a mapping, and not to placeholders, which merge every entity of one type by design.

2.5 The Attacks

Frequency matching aligns surrogate frequencies with real-name frequencies. Published attacks on pseudonymised free text target the identifiers a system missed rather than the surrogate distribution it produced, as the parrot attack does [5], so we transfer the alignment model from attacks on encoded identifiers in record linkage and disclosure-risk work [23, 40]. Two priors run over one observation, following Kuzu et al., Niedermeyer et al. and Vatsalan et al. [17,24,39]: a corpus-internal prior granting perfect knowledge, an oracle upper bound, and a public name-frequency list, the realistic adversary.

The attacker runs its own detector on the released text, so it can identify names that the defender failed to replace. All detected surfaces remain in the evaluation denominator, including ambiguous and unresolved cases [8]. If the same surrogate represents k different entities, a correct recovery receives a score of $1/k$ rather than being excluded. The attacker abstains when the score margin is too small, using $(\max - \max_2)/\sigma < \varphi$, where max and $\max_2$ are the best and second-best candidate scores, σ their spread across candidates, and φ the abstention threshold, all computed from its own candidate scores [22]. We therefore report both the fraction of queries answered and c@1, which rewards correct answers while allowing justified abstention [30], together with the chance level implied

Table 2: Operating points on both error rates. Sensitivity is over the identifier types the conditions replace, specificity over every other token. Cost is seconds per document; the rule is ∪ union, ∩ intersection, v at-least-k vote, s a single detector.

	CARDIO:DE			TAB			OntoNotes			Enron		
Operating point	Sens.	Spec.	Cost	Sens.	Spec.	Cost	Sens.	Spec.	Cost	Sens.	Spec.	Cost
max. sensitivity	0.9919	0.9670	$57.633^{\cup}$	0.9253	0.8986	$14.210^{\cup}$	0.7466	0.9763	$9.450^{\cup}$	0.9968	0.7745	$25.557^{\cup}$
fast sensitivity	0.8947	0.9769	$0.301^{\cup}$	0.8926	0.9242	$0.219^{\cup}$	0.6843	0.9731	$5.376^{\cup}$	0.9222	0.7420	$0.057^{\cup}$
max. specificity	0.5230	0.9997	116.875^{v}	0.5014	0.9959	98.110^{v}	0.5034	0.9960	$0.103^{\cup}$	0.5292	0.9448	$20.931^{\cap}$
fast specificity	0.7807	0.9862	0.263^{s}	0.5198	0.9934	0.117^{s}	0.5034	0.9960	$0.103^{\cup}$	0.6920	0.8781	0.026^{s}

by the candidate set [33]. Realistic surrogates are intended to make residual identifiers difficult to distinguish from replaced ones [4]; the public-prior attack tests this property directly. Varying the prior's size and publication date then quantifies the value of additional attacker knowledge [3].

Context linkage ranks identities by cosine similarity between surrounding-word profiles; *learned linkage* fits the same task on labelled, entity-disjoint examples. Both use identical document-disjoint folds, and only the labels of held-out queries are hidden: those people stay in the complete candidate list with every distractor, so neither attack is told that a query has a match, and their difference measures learning rather than gallery size. Cross-document identity permits this on Enron and CARDIO:DE but not on TAB or OntoNotes.

LLM candidate ranking gives a language model (*gpt-oss-120B*) one marked mention and ten candidate names, with or without auxiliary text about each candidate from the released condition. Because conditions B and C replace identical spans, B can be rewritten into C, so we report this attack on B as presented, on B after that rewrite, and against a ranker told the scheme. The one-in-ten closed world measures an attacker holding a shortlist, not population re-identification. A *dictionary lookup* would hash candidate names; it is defined but not run, because no condition here is unkeyed.

3 Results

3.1 Detection

Table 2 gives the cost–quality front for our detector triplet ensembles, with sensitivity the primary criterion for protection. Maximum-sensitivity configurations reach 0.9919 on CARDIO:DE, 0.9253 on TAB, 0.7466 on OntoNotes, and 0.9968 on Enron. The faster alternatives retain 0.8947, 0.8926, 0.6843, and 0.9222 sensitivity, respectively, while sharply reducing cost; for example, CARDIO:DE falls from 57.633 s to 0.301 s and Enron from 25.557 s to 0.057 s per document. The OntoNotes specificity rows coincide because the 10 % band admits all 414 candidates.

Within OntoNotes, language effects depend on what is detected. Under the recommended ensemble, sensitivity is 0.9890 on English, 0.8680 on Chinese and 0.6137 on Arabic; pooled sensitivity is 0.8011. Arabic supplies 45 % of identifier

Table 3: Leakage from the recommended 13-detector union; all columns concern people. *Sens.* is person sensitivity and *Spec.* the share of non-identifier tokens left alone: neither is readable without the other. *Exp.* is the risk that a person is still named in a given document after processing, *Frequency* the identities named from frequency alone with a public list and, bracketed, with the corpus's own distribution (an upper bound, not a risk). *Linkage* is the percentage of queries ranking the correct person first, over the complete candidate list, against chance 1/gallery.

Corpus	Sens.	Spec.	Exp. (%)	Frequency	Context link.	Learned link.	Chance
CARDIO:DE	0.9998	0.8686	0.05	0 (1)	0	0.71 ± 1.60	1/207
Enron	0.9906	0.6235	1.60	0 (0)	0.93 ± 0.10	3.94 ± 0.30	1/3697
TAB	0.9958	0.8504	0.83	0 (1)	no links → not measurable		–
OntoNotes	0.9352	0.9318	5.43	0 (1)	no links → not measurable		–

tokens from 7 % of documents, so it dominates the pooled result while remaining the hardest case. Person sensitivity transfers better at 0.9913, 0.9315 and 0.8783, pooling to 0.9352. The thirty-seven-point spread therefore lies in the other identifier classes rather than the people targeted by the attacks.

3.2 Utility

At the high-sensitivity operating point, utility depends strongly on replacement form. On CARDIO:DE, named-entity agreement is 0.611 ± 0.082 with surrogates but only 0.004 ± 0.008 with placeholders. The latter largely reflects the frozen recognizer rather than semantic degradation: it never tags `[PERSON]` as a name, so every replaced mention is counted as a disagreement. Surrogates therefore preserve token-level named-entity utility substantially better than placeholders. The same ordering holds on the other three corpora: named-entity agreement under the permissive rule is 0.478 on TAB, 0.295 on OntoNotes and 0.330 on Enron, rising to 0.898, 0.888 and 0.754 under the precise rule.

The other two tasks retain their original-text baselines and support paired comparisons. Medication extraction scores 0.371 ± 0.182 on unmodified text. Under the precise rule, surrogate replacement yields 0.364 ± 0.189, with a paired difference of -0.008 ± 0.178 that is consistent with zero ($q = 0.80$). Under the permissive rule, performance drops to 0.346 ± 0.181, corresponding to a paired difference of -0.020 ± 0.182 ($q = 5.6 \times 10^{-3}$). Section classification scores 0.749 ± 0.093 on unmodified text. Surrogates reduce this score by 0.072 ± 0.086, whereas placeholders reduce it by 0.038 ± 0.089; both effects are significant ($q = 1.6 \times 10^{-37}$ and $q = 2.6 \times 10^{-14}$; rank-biserial correlations -0.846 and -0.531). Thus, typed placeholders preserve more utility at the document level, whereas surrogates are required to preserve token-level utility.

3.3 Leakage

Table 3 reports the 13-detector union. This frequency attack with the public name list recovers zero identities by alignment across the four corpora. The

names it did get right were ones the detector missed and the release printed in clear text, so that count measures defender sensitivity rather than adversarial inference. The corpus distributions, unavailable to a real attacker, align at most one identity. Raising the abstention threshold from $\varphi = 0$ to $\varphi = 0.05$ leaves 11 of 25,677 CARDIO:DE queries and zero elsewhere; alignment c@1 is zero at every threshold.

Linkage requires cross-document identity and must be read against unmodified text and chance. On Enron, fixed similarity ranks the correct person first for 71.98 % of queries on unmodified text and 0.93 ± 0.10 % after replacement. The latter is 34 times the 1/3697 chance rate but two orders of magnitude below the unmodified-text rate. Entity-disjoint training raises recovery to 3.94 ± 0.30 %, four times the fixed attack and 146 times chance, so a trained adversary carries the residual risk. CARDIO:DE has 207 attacker's reference identities; fixed linkage recovery is zero, and learned linkage 0.71 ± 1.60 % over 138 queries against chance 1/207 is consistent with guessing. Across the tested Enron configurations, linkage rises as person sensitivity falls. Agreement between two detectors lowers sensitivity to 0.8310, raises fixed linkage fivefold to 5.09 % and increases clear-text names from 152 to 1,816; intersection yields 0.0443 sensitivity and 25.51 % linkage.

The third attack assigns different risks to the replacement forms. Ranking one marked mention against ten candidates, the model falls from 0.995 on unmodified TAB to 0.110 with surrogates, near 0.100 chance, but reaches 0.340 with typed placeholders; OntoNotes (0.290 versus 0.630) orders likewise. Only CARDIO:DE (0.085 versus 0.120) stays at chance level for both categories. This is consistent with source-text recall, not stronger surrogate protection: a neutral placeholder permits retrieval of a memorized person, whereas a surrogate supplies a conflicting name and suppresses that candidate. Accordingly, C is highest on widely replicated OntoNotes, intermediate on public TAB, and at chance on restricted CARDIO:DE. Yet both forms replace identical spans. Replacing each surrogate with `[PERSON]` reproduces C character for character on all 1,268 TAB and 5,994 OntoNotes documents.

Sensitivity alone omits exposure units; each denominator answers a different question. The 13-detector union leaves one CARDIO:DE person mention of 2,800 in clear text, in one letter of 400 and for one patient of 270; over every identifier type, 9.5 % of letters carry something and 13.3 % of patients do. TAB leaves a name in 47 of 1,268 judgments and OntoNotes in 790 of 5,994 documents. On Enron the risk that a person is still named in a given message is 1.6 %, while 5.9 % of messages carry an identifier of some type.

The key normaliser both merges and splits identities: on Enron, ORGANISATION surrogates collide at 17.3 % and drift at 48.6 %, and PERSON drift reaches 37.7 % against 5.0 % on CARDIO:DE. The lower CARDIO:DE figure reflects constructed recurrence with fewer natural variants rather than generalisation. Drift is a utility cost as well as a mapping failure, because it breaks the longitudinal link that stable pseudonyms exist to preserve. It is a property of the key normaliser, and a stricter one reduces it at the price of more collisions.

4 Discussion

For releases prioritising leakage reduction, the measurements support the large union, whose CARDIO:DE specificity is 0.8686. Requiring detector agreement trades sensitivity for specificity. On Enron, two votes lower person sensitivity from 0.9906 to 0.8310 and raise fixed linkage fivefold. Across these tested configurations, lower sensitivity coincides with greater name exposure and linkage.

Attack results depend on adversarial knowledge and denominator. The public prior measures this frequency attack with one available source; its contrast with the corpus-internal upper bound is the frequency result. Corpus-level scores should accompany document, case and language results.

For TAB and OntoNotes, a scheme-aware adversary can derive the placeholder release from the surrogate release. Replacement form still affects utility. Telling the ranker the scheme raises recovery from 0.110 to 0.165 on unchanged text. In the tested tasks, surrogates preserve token-level performance and placeholders more document-level utility.

The pre-registered prediction for these attacks, priors, corpora and ensemble was that increasing detector sensitivity would increase frequency-matching recovery and decrease context-linkage recovery. Both components were rejected: frequency alignment yielded zero identities at every tested sensitivity, while both linkage attacks decreased as sensitivity increased. Missed identifiers limited both measured attacks.

Linkage was measurable only on Enron and CARDIO:DE because TAB and OntoNotes scope every entity identifier to one document; their zero linkage values arise from the annotation and understate real risk, which limits the measurement rather than the attacks. CARDIO:DE names were inserted into an already de-identified release, and linkage recovers only 2.73 % on unmodified text, so its near-zero post-replacement values reflect corpus properties rather than protection effectiveness. The candidate-ranking attack presents ten names and therefore represents an adversary with a shortlist, not population search.

5 Conclusion

Safe sharing of clinical data needs both modalities protected: a guarded image does not compensate for a report that names the patient. This study measures the text half; the image half is a separate literature. Artificial intelligence makes both possible at scale, but its failures must be measured where data are released. For text we priced detection, utility and re-identification on the same documents. The 13-detector ensemble replaces almost every person mention; linkage is reduced by about two orders of magnitude from unmodified text but remains 3.94 % for a trained e-mail attacker. Detector configuration governs utility more than replacement form, and sensitivity understates exposure per document, case and language. There is therefore no single anonymisation score: a release should state its operating point, per-language errors, surviving cases, attacker knowledge and stability. Our operating points and measurements are released with acceptance of this paper.

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